

What matter(s) between galaxies, 06.06.19

Probing the IGM thermal state with Ly- α absorption across cosmic time

Jamie Bolton

With thanks to:

Fahad Nasir, Elisa Boera, George Becker, Vid Irsic, Ewald Puchwein,
Matteo Viel, Martin Haehnelt, Tae-Sun Kim



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Key question

- Q3: What determines the thermal and ionisation state of the IGM/CGM?

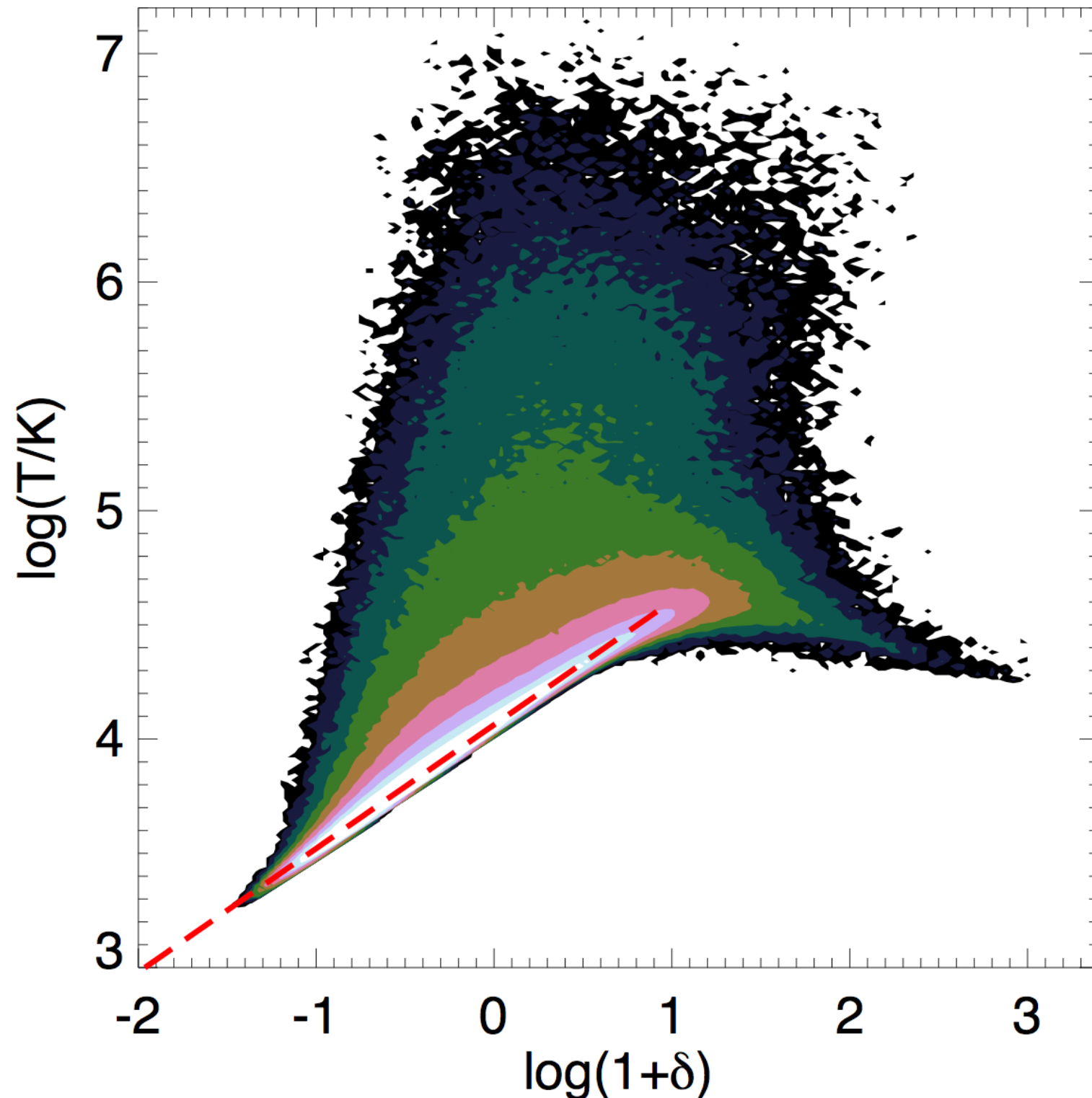
Key question

- Q3: What determines the thermal and ionisation state of the IGM/CGM?
- IGM temperature and COS line widths at $z=0.1$
- Modelling effect of reionisation on IGM at $z>5$

Brief motivation

- Indirect probe of (H-I and He-II) reionisation timing;
- Sensitive to spectral shape of ionising background (i.e. AGN vs stars as sources);
- Important “nuisance parameter” for dark matter coldness and/or cosmology constraints from forest.

The temperature-density relation



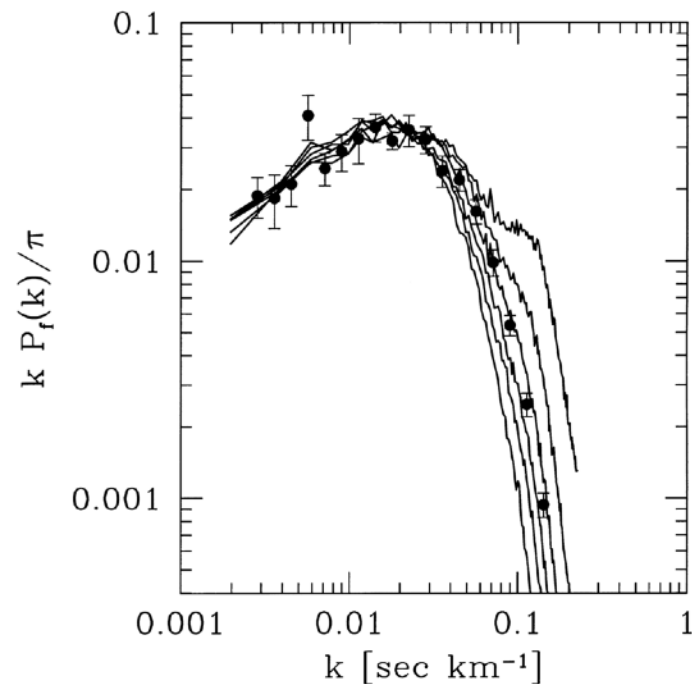
Power-law relationship
between temperature
and density, $\gamma \sim 1.0-1.6$

$$T = T_0(1 + \delta)^{\gamma-1}$$

Hui & Gnedin (1997),
McQuinn & Upton
Sanderbeck (2016)

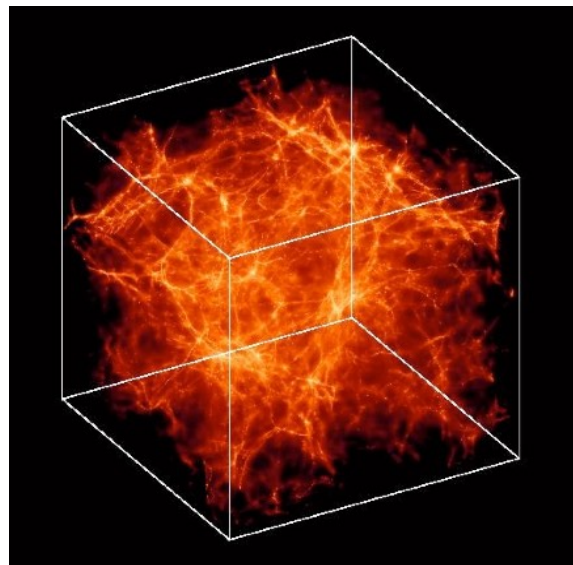
Measuring the temperature

Transmitted flux statistics

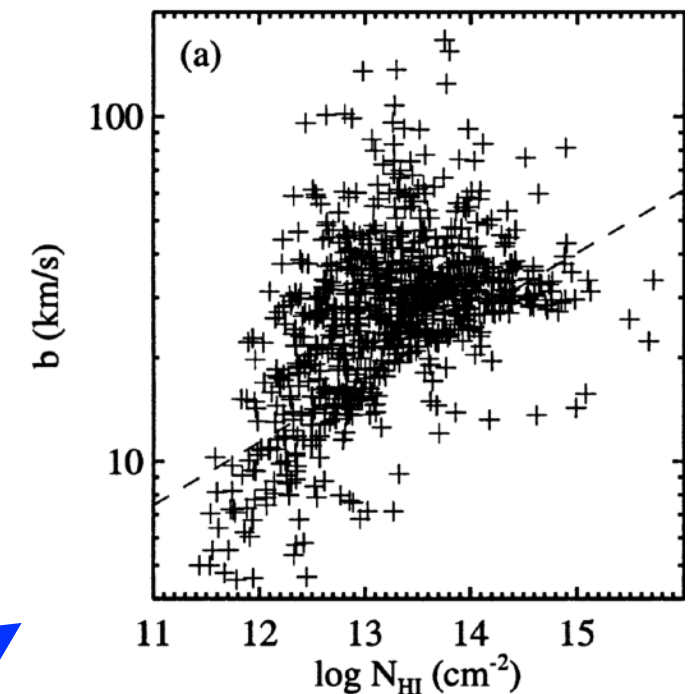


[Zaldarriaga et al. \(2001\) \[P\(k\)\]](#)
[Theuns et al. \(2002\) \[wavelets\]](#)
[Bolton et al. \(2008\) \[PDF\]](#)
[Viel et al. \(2009\) \[P\(k\)+PDF\]](#)
[Lidz et al. \(2010\) \[wavelets\]](#)
[Becker et al. \(2011\) \[curvature\]](#)
[Garzilli et al. \(2012\) \[wavelets\]](#)
[Calura et al. \(2012\) \[PDF\]](#)
[Boera et al. \(2014, 2016\) \[curvature\]](#)
[Lee et al. \(2015\) \[PDF\]](#)
[Rorai et al. \(2017\) \[PDF\]](#)
[Walther et al. \(2019\) \[P\(k\)\]](#)

IGM model



Line decomposition



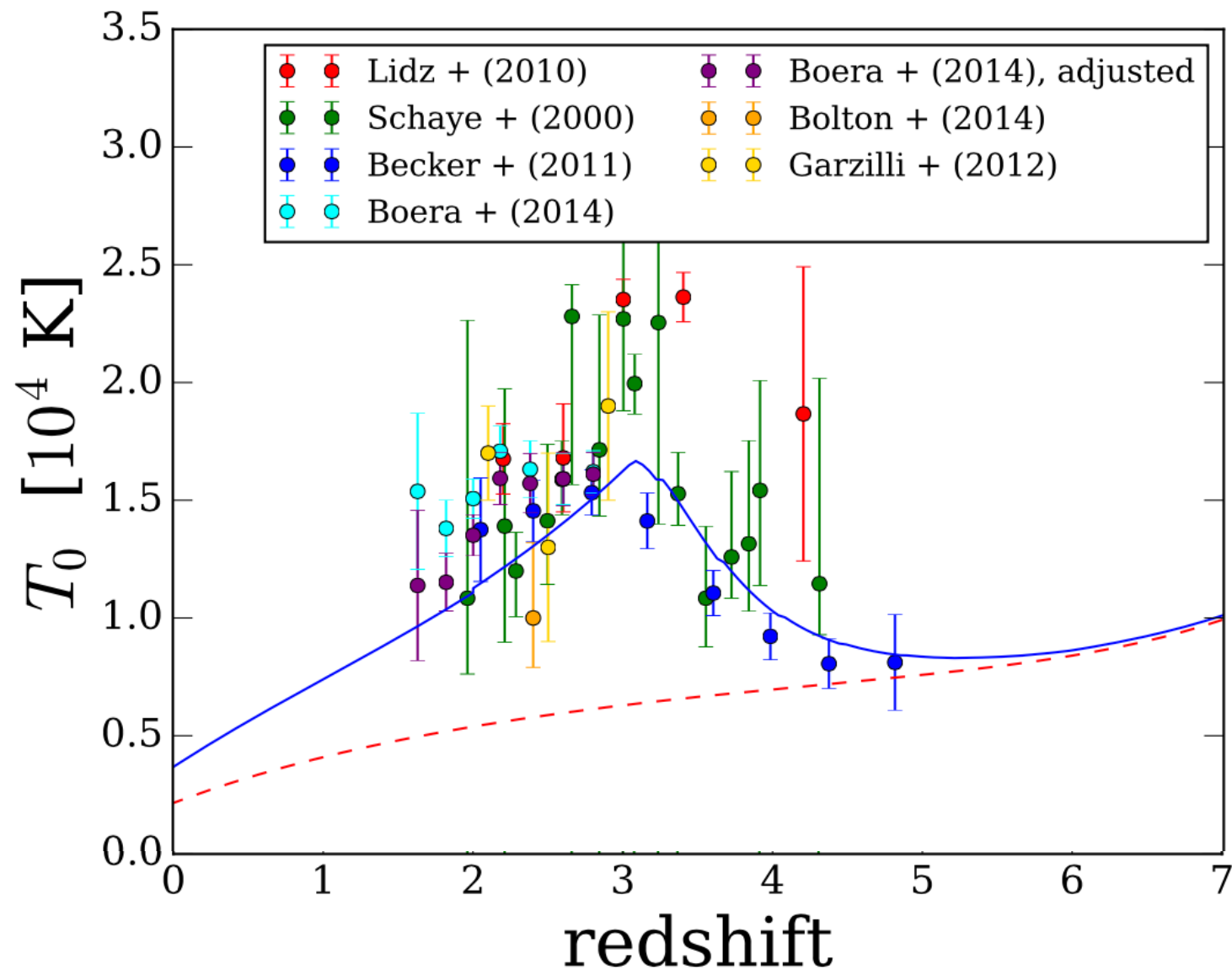
[Schaye et al. \(2000\)](#)
[Ricotti et al. \(2000\)](#)
[Bryan & Machacek \(2000\)](#)
[McDonald et al. \(2001\)](#)
[Bolton et al. \(2012, 2014\)](#)
[Rudie et al. \(2012\)](#)
[Rorai et al. \(2018\)](#)
[Hiss et al. \(2018\)](#)
[Telikova et al. \(2018\)](#)

$$T_0, \gamma$$

$$(\sigma_J)$$

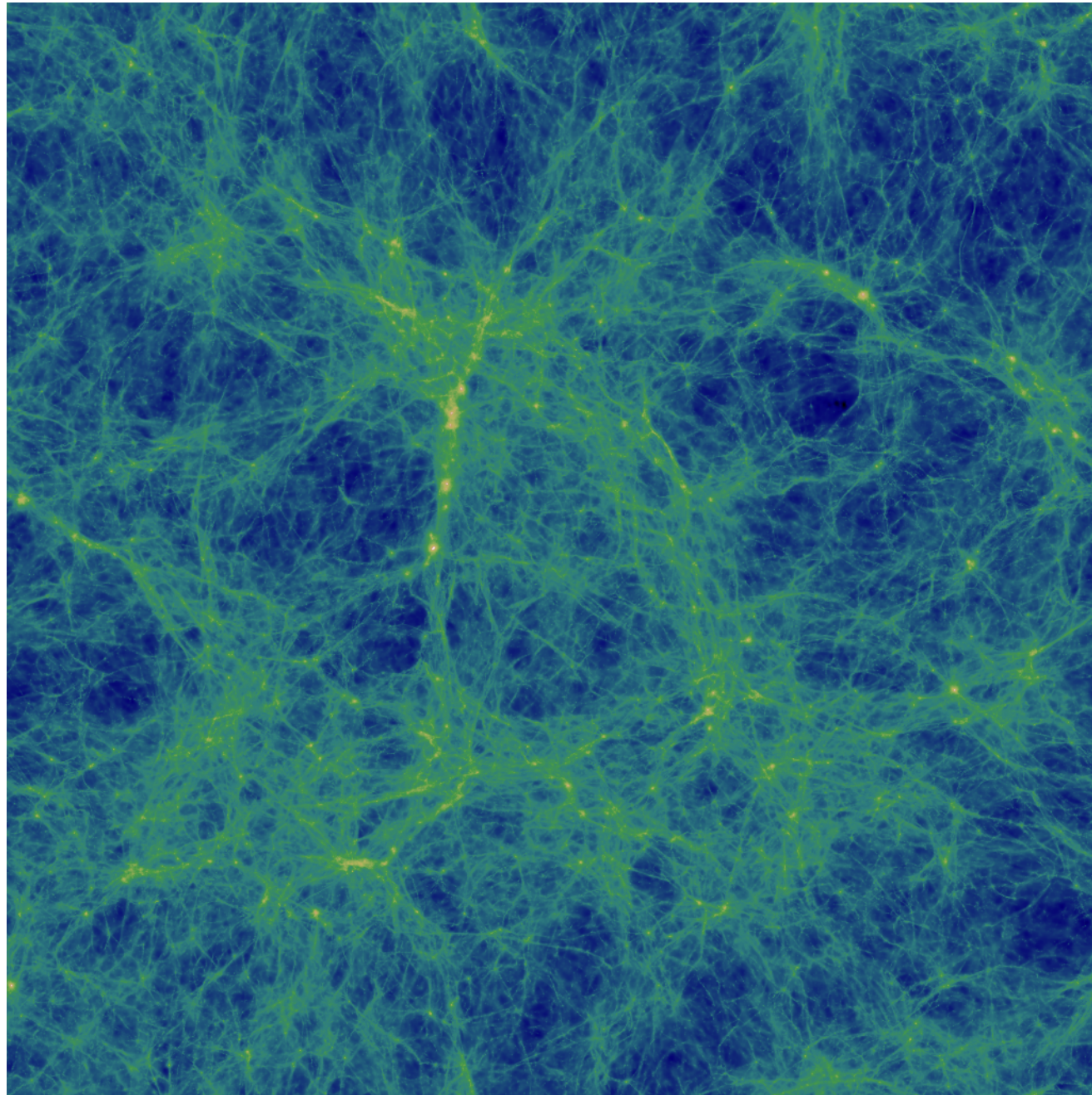
The temperature evolution ($\Delta=0$)

Data mainly at $2 < z < 4$. Overall picture is where He-II reionisation heats IGM by $\sim 7000\text{K}$ at $z < 4.5$.



[Upton Sanderbeck et al. \(2016\)](#), see also [Puchwein+18](#), [D'Aloisio+18](#) for discussion in context of AGN reionisation models

The Sherwood simulation suite

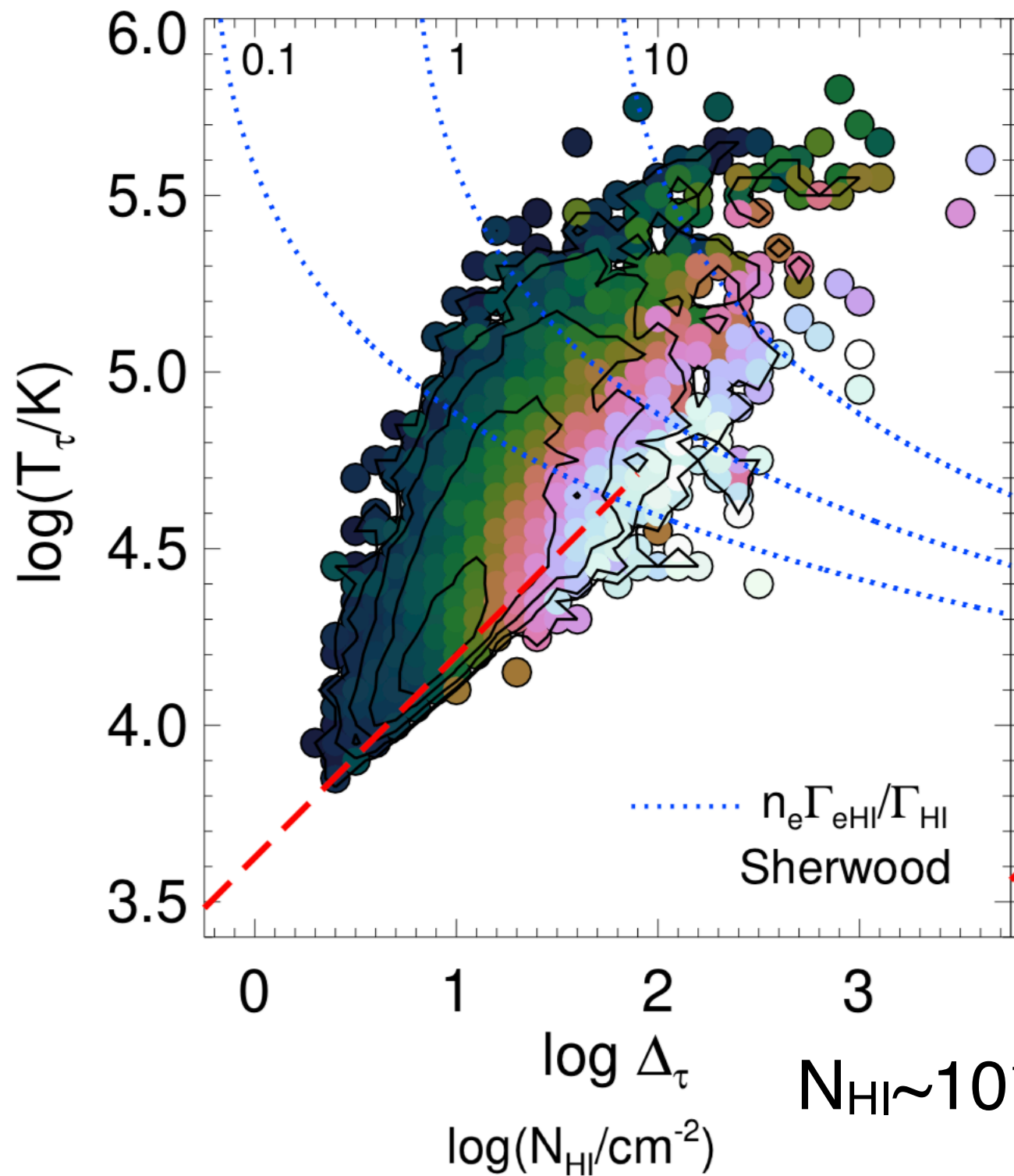


- 40-160 Mpc/h with 2×2048^3 particles, $M_{\text{gas}} = 10^5 M_{\odot}/h$; Planck cosmology; hydrodynamical IGM calculations with P-Gadget-3; $\sim 70\text{Tb}$ data
- CDM and WDM (3.5 keV thermal relic) runs, varying reionisation histories;
- Stellar and AGN feedback (for selected models) follows Puchwein & Springel (2013).

40 comoving Mpc/h at $z=2.8$

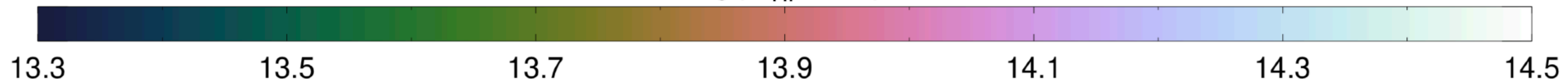
Bolton et al. (2017)

Ly- α forest at $z=0.1$

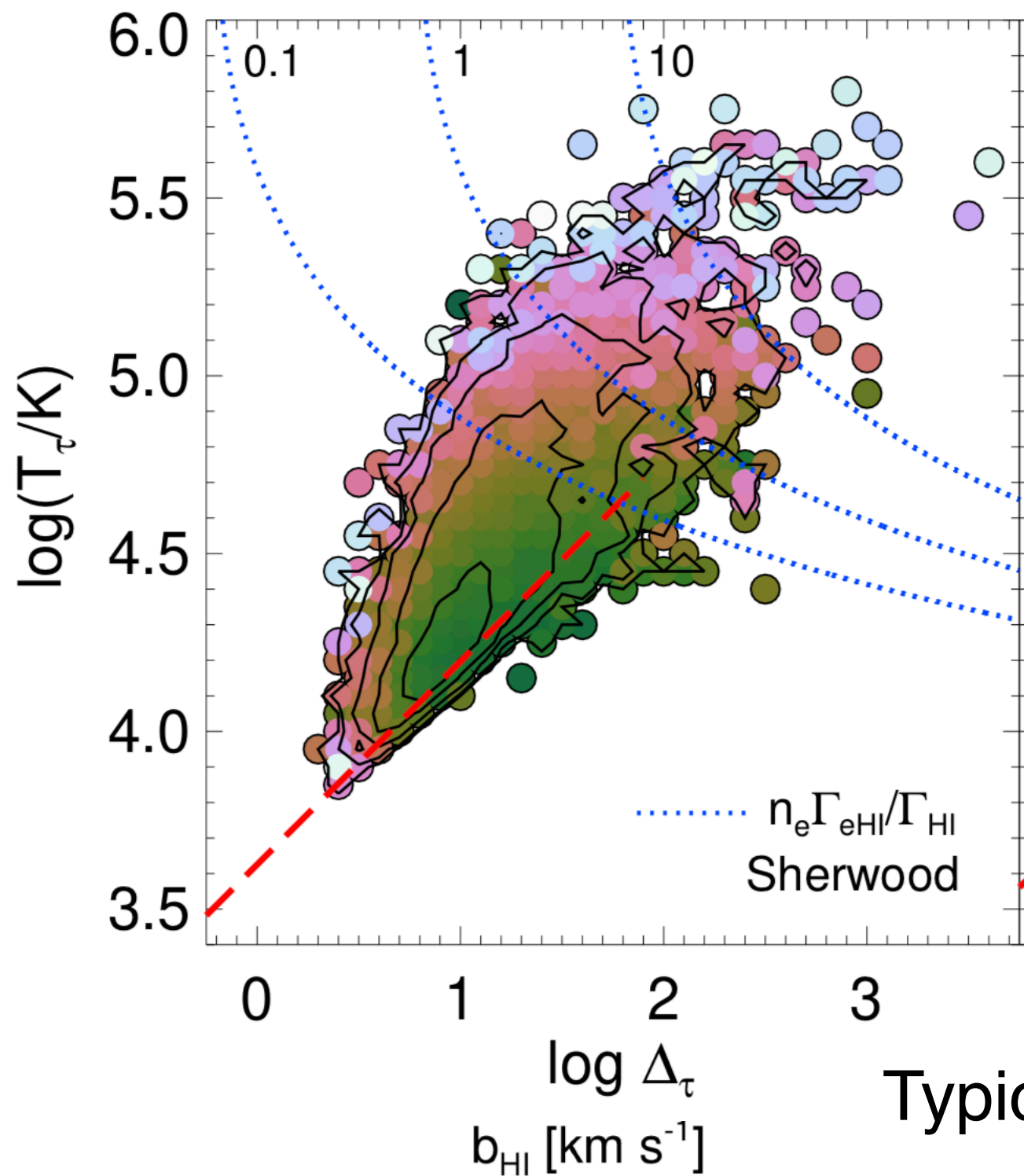


Lines obtained by
Voigt profile fitting
mock COS spectra,
FWHM $\sim$ 17 km/s

$N_{\text{HI}} \sim 10^{13.5} \text{ cm}^{-2}$ has $\Delta \sim 10$

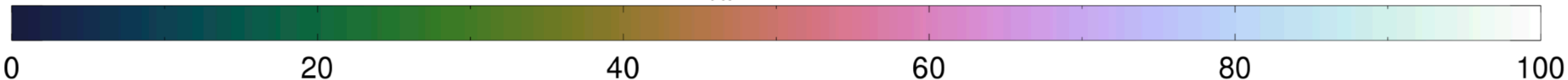


Ly- α forest at $z=0.1$

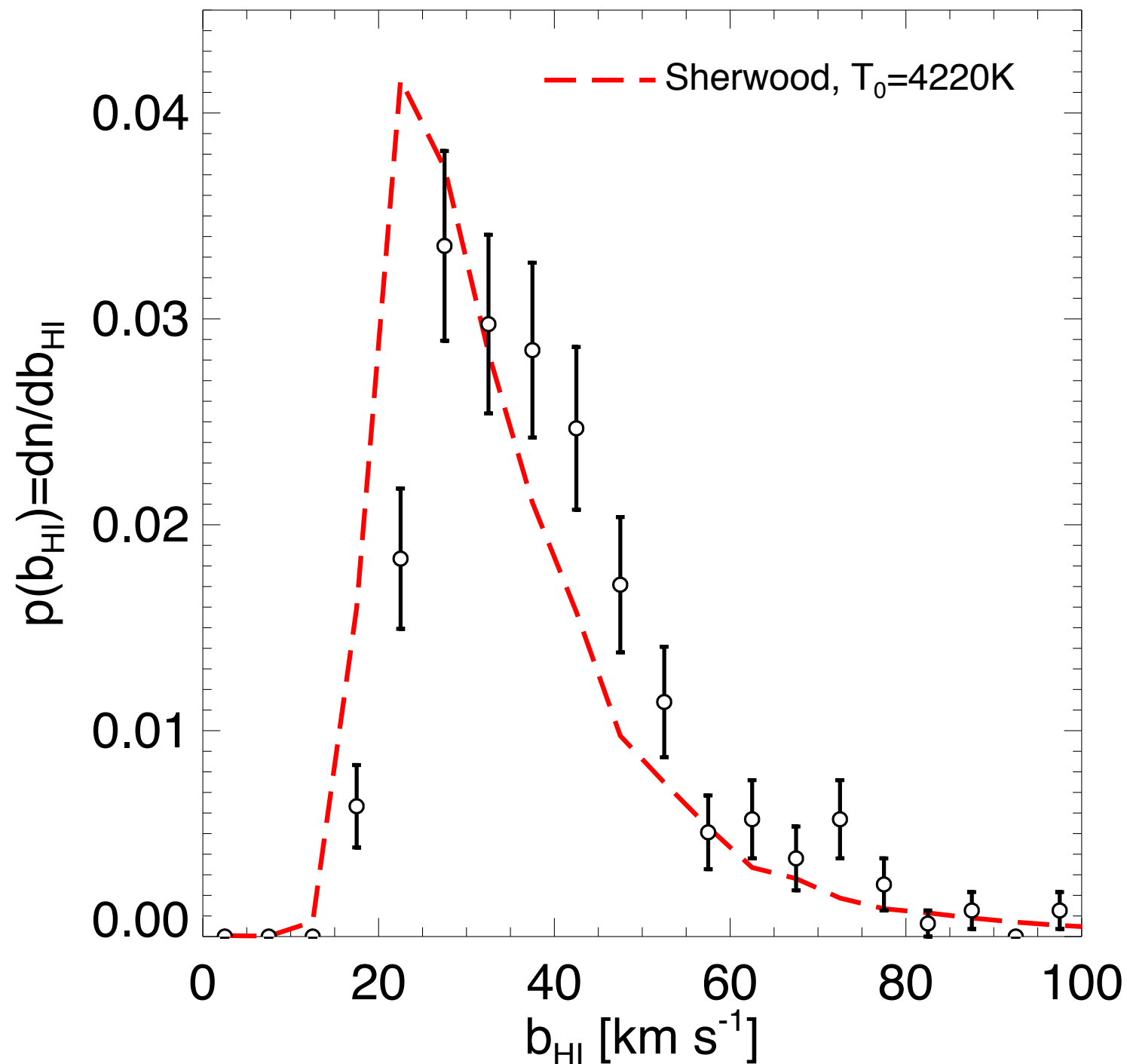


Lines obtained by
Voigt profile fitting
mock COS spectra,
FWHM ~ 17 km/s

Typical $b_{\text{HI}} \sim 20\text{-}30$ km s $^{-1}$



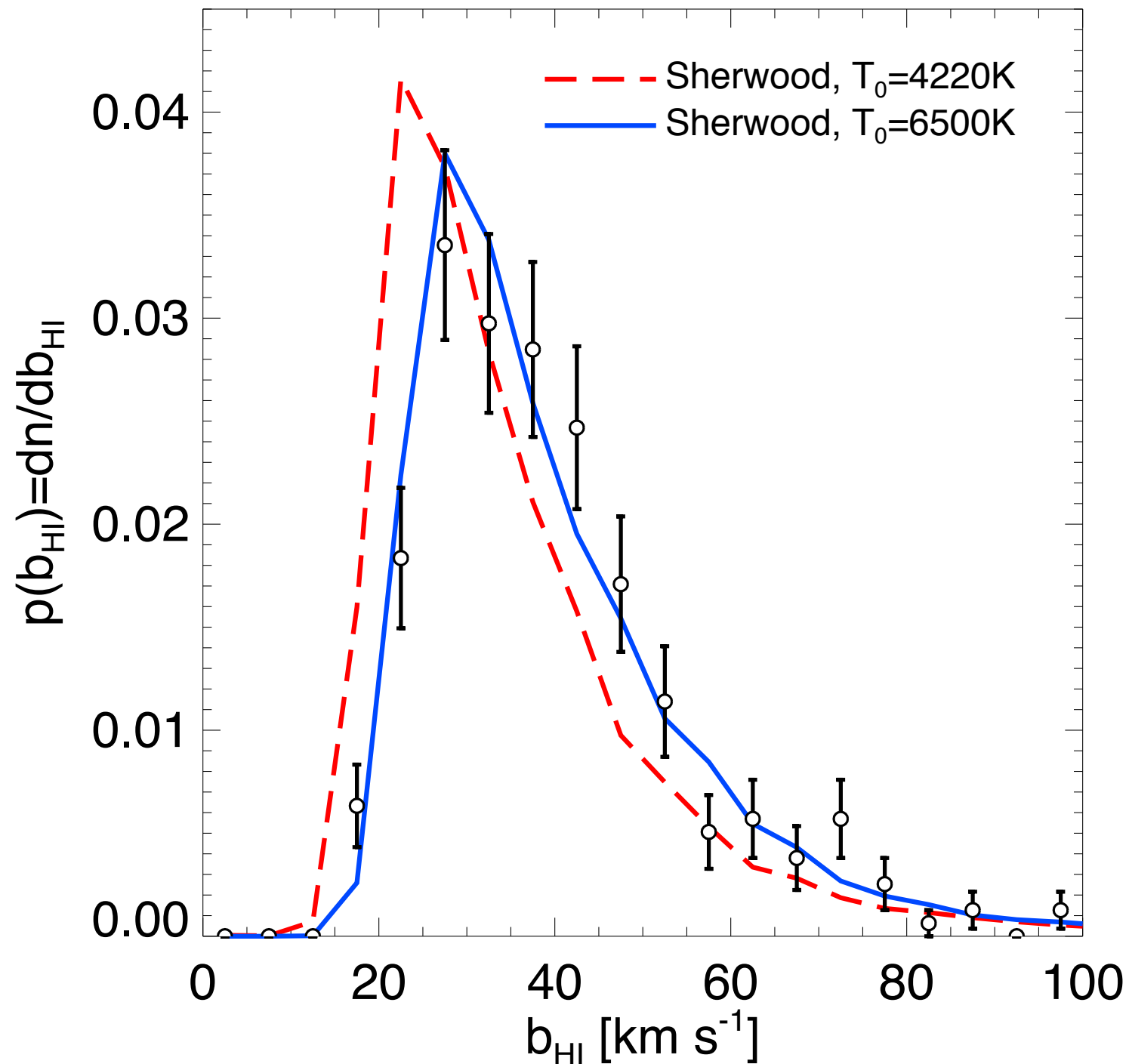
Line widths at $z=0.1$



- Ly- α line fits at $0 < z < 0.2$; from 44 COS spectra;
- Forward model Ly- α spectra and fit Voigt profiles with VPFIT (use COS LSF, S/N..)
- Simulated line widths too narrow at $z=0.1$.

Viel et al. (2017);
Nasir et al. (2017)

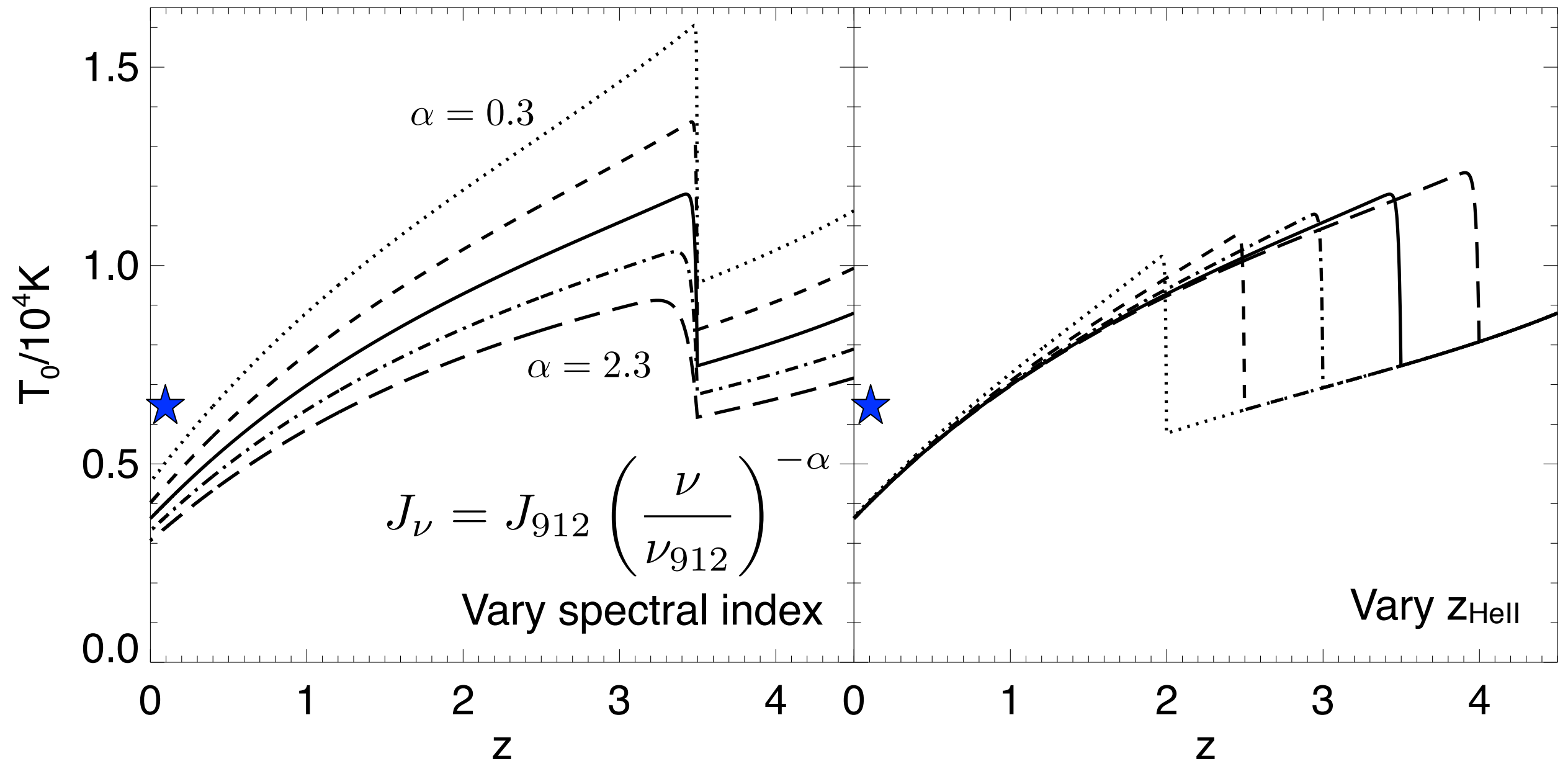
Line widths at $z=0.1$



- Can improve agreement by boosting IGM temperature. Is this physically motivated?

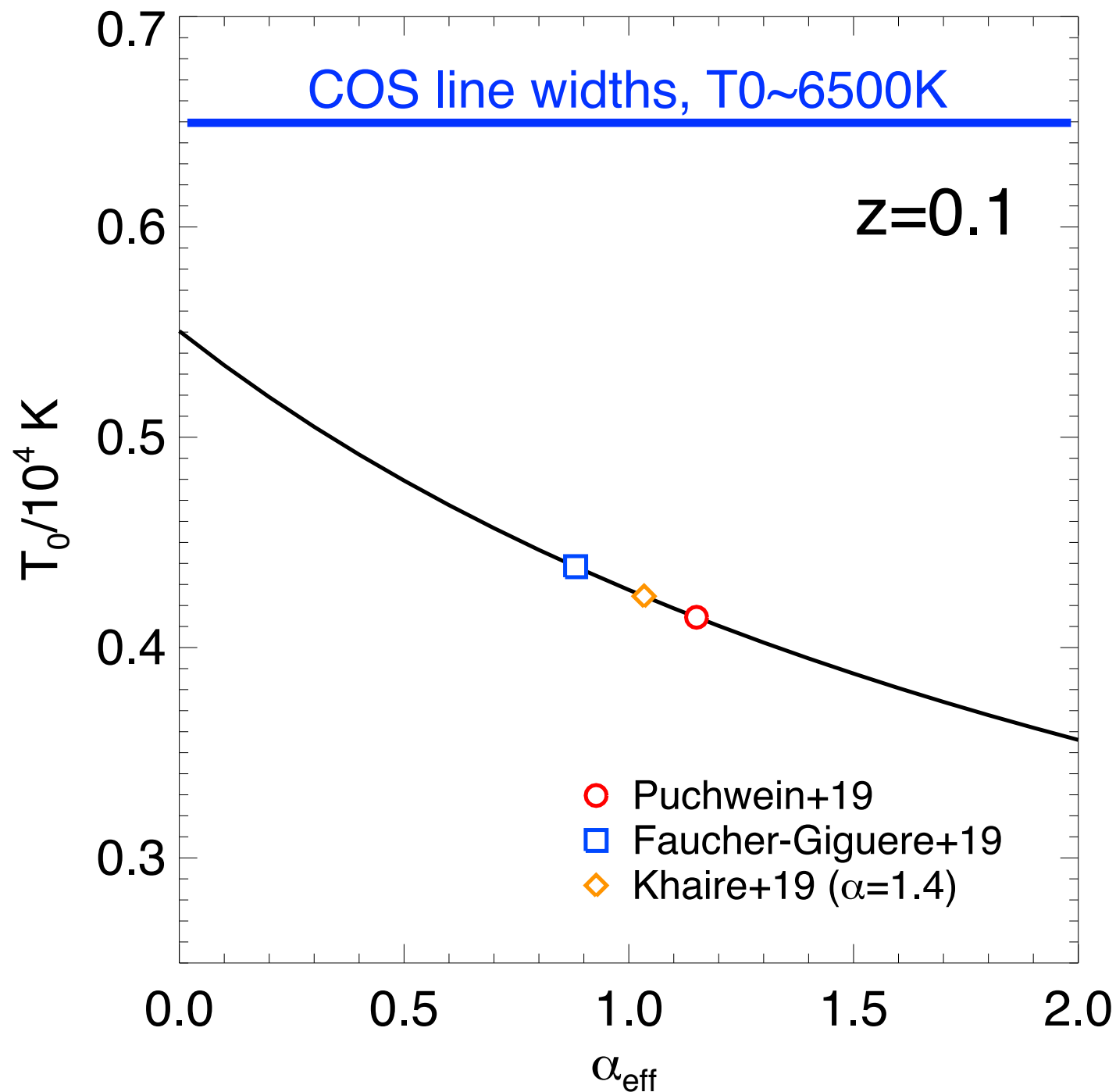
Viel et al. (2017);
Nasir et al. (2017)

Additional photo-heating?



Evolution well understood: cooling to $z=0$ due to adiabatic expansion.
See also [Hui & Haiman \(2003\)](#).

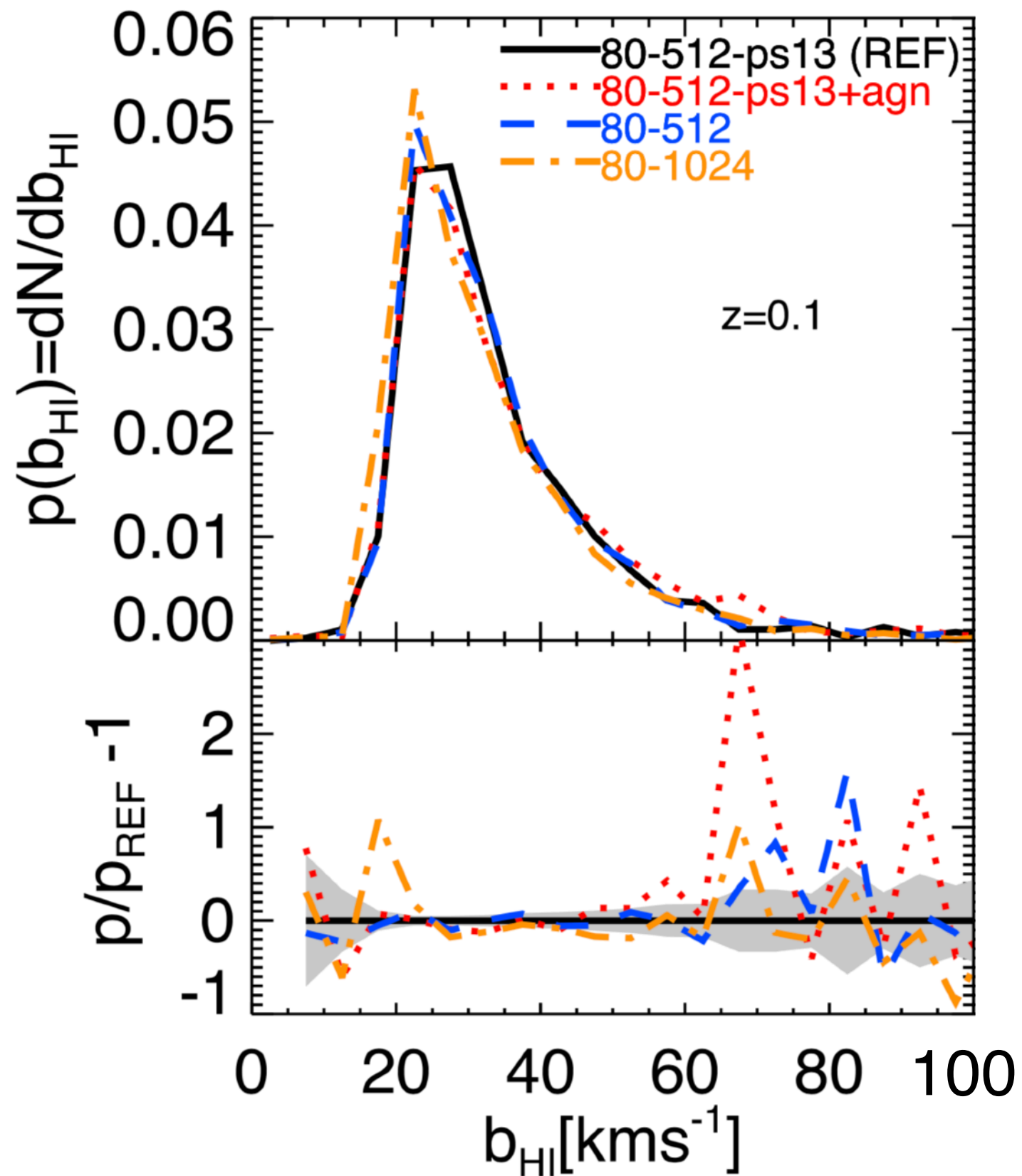
Additional photo-heating?



- UVB models produce similar temperatures from photo-heating;
- (Unreasonably) hard UVB spectrum required to match line widths.

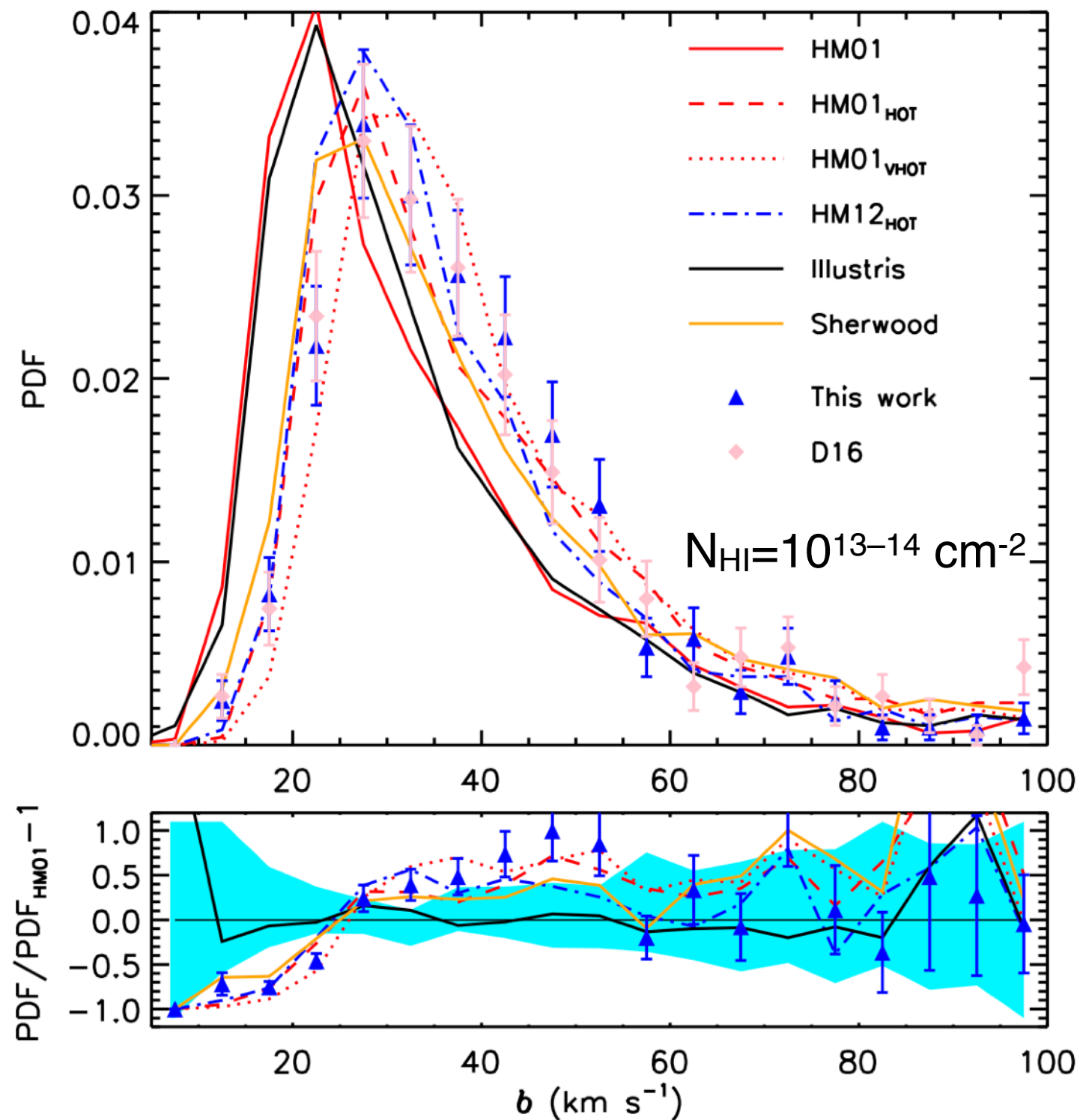
$$J_\nu = J_{912} \left(\frac{\nu}{\nu_{912}} \right)^{-\alpha}$$

AGN feedback?



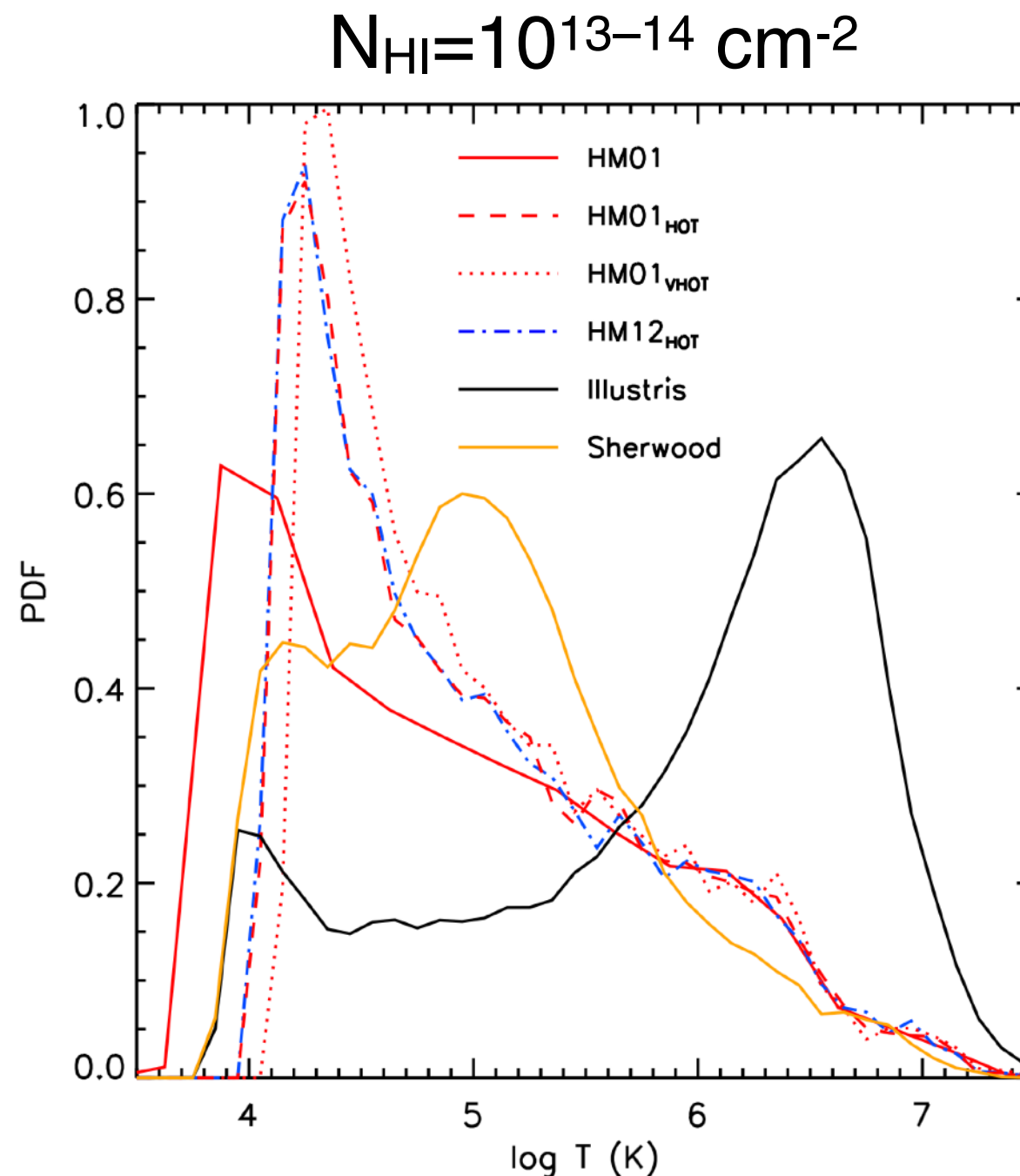
Viel et al. (2017);
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AGN feedback?



Viel et al. (2017);
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AGN feedback?

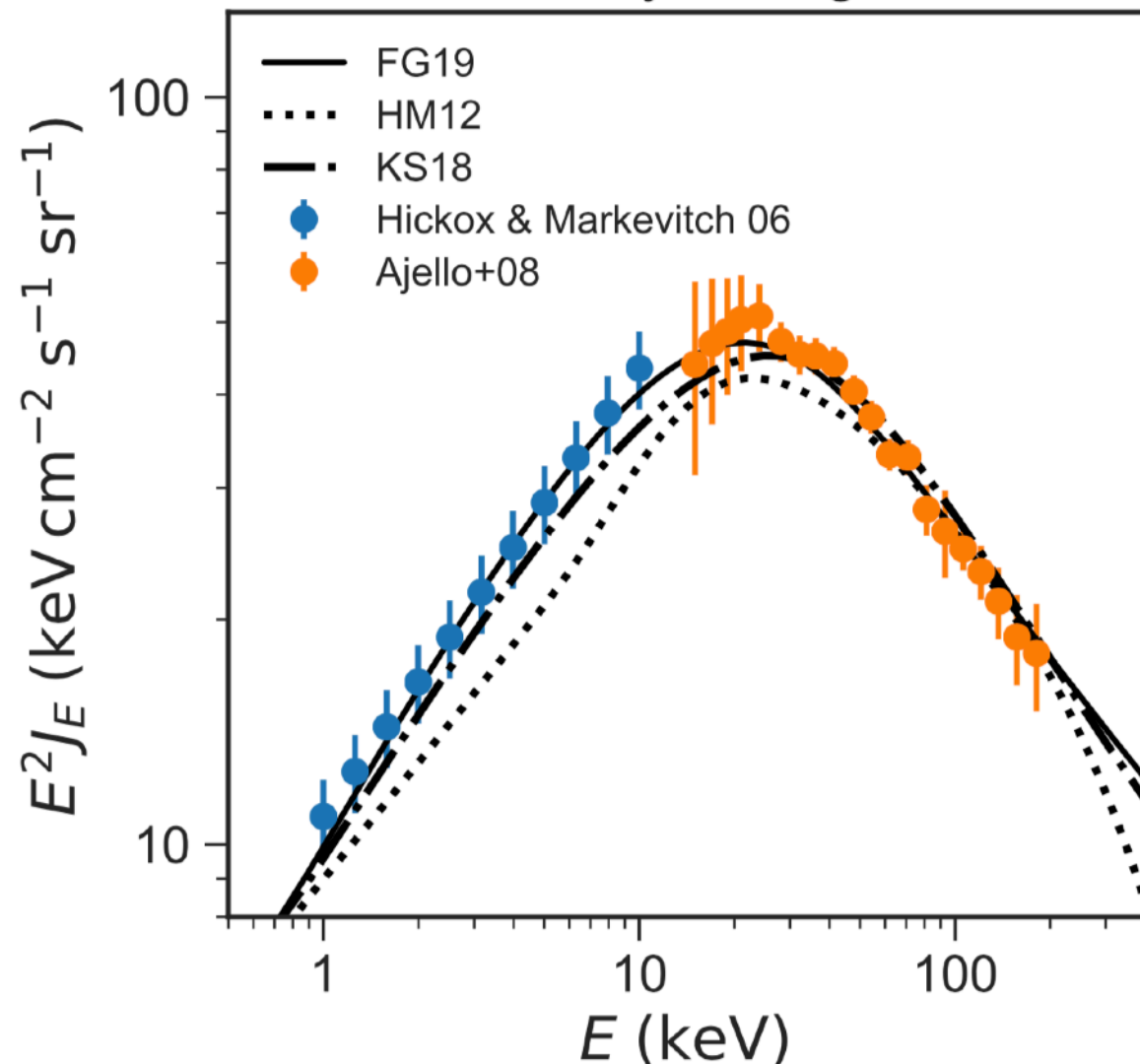


Viel et al. (2017)

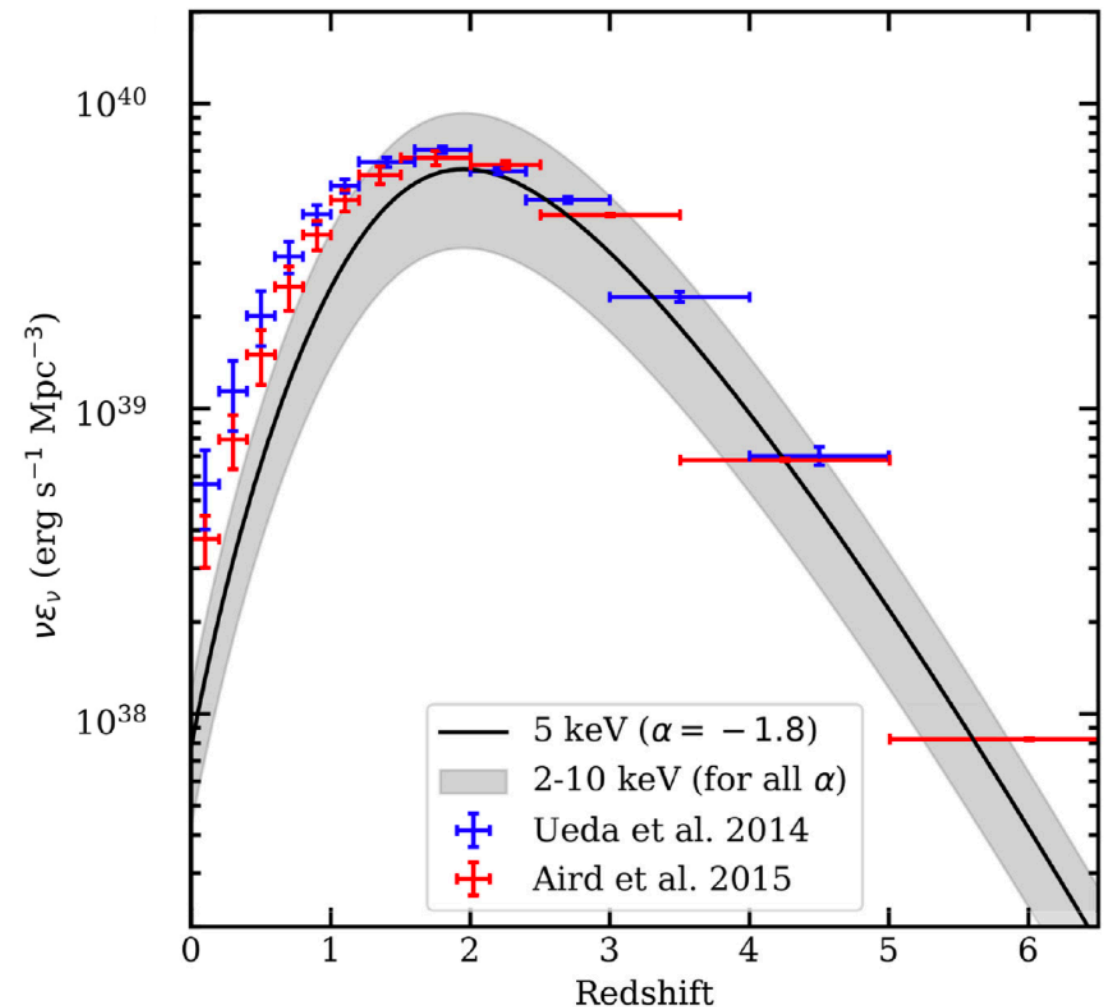
Illustris AGN feedback produces a large volume filling factor of $T > 10^6 \text{ K}$ gas at $z \sim 0$ (WHIM). Collisionally ionised $\rightarrow$ no Lyman-alpha absorption!

Compton heating by X-rays?

Local X-ray background



Faucher-Giguère (2019)

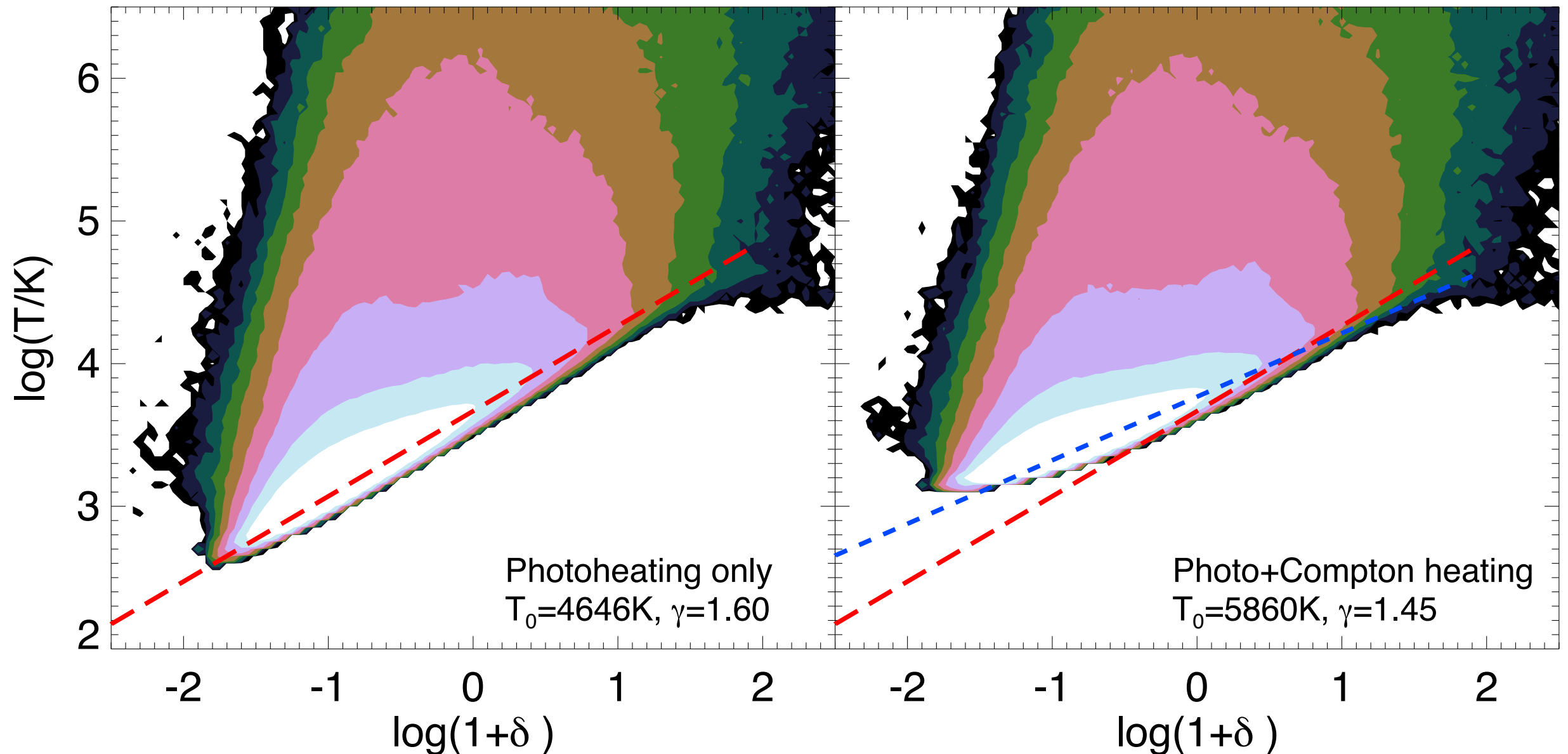


Khaire et al. (2019)

$$H_C = \frac{4\pi n_e}{m_e c^2} \int_0^\infty J_\nu \sigma_\nu (h\nu - 4k_B T) d\nu$$

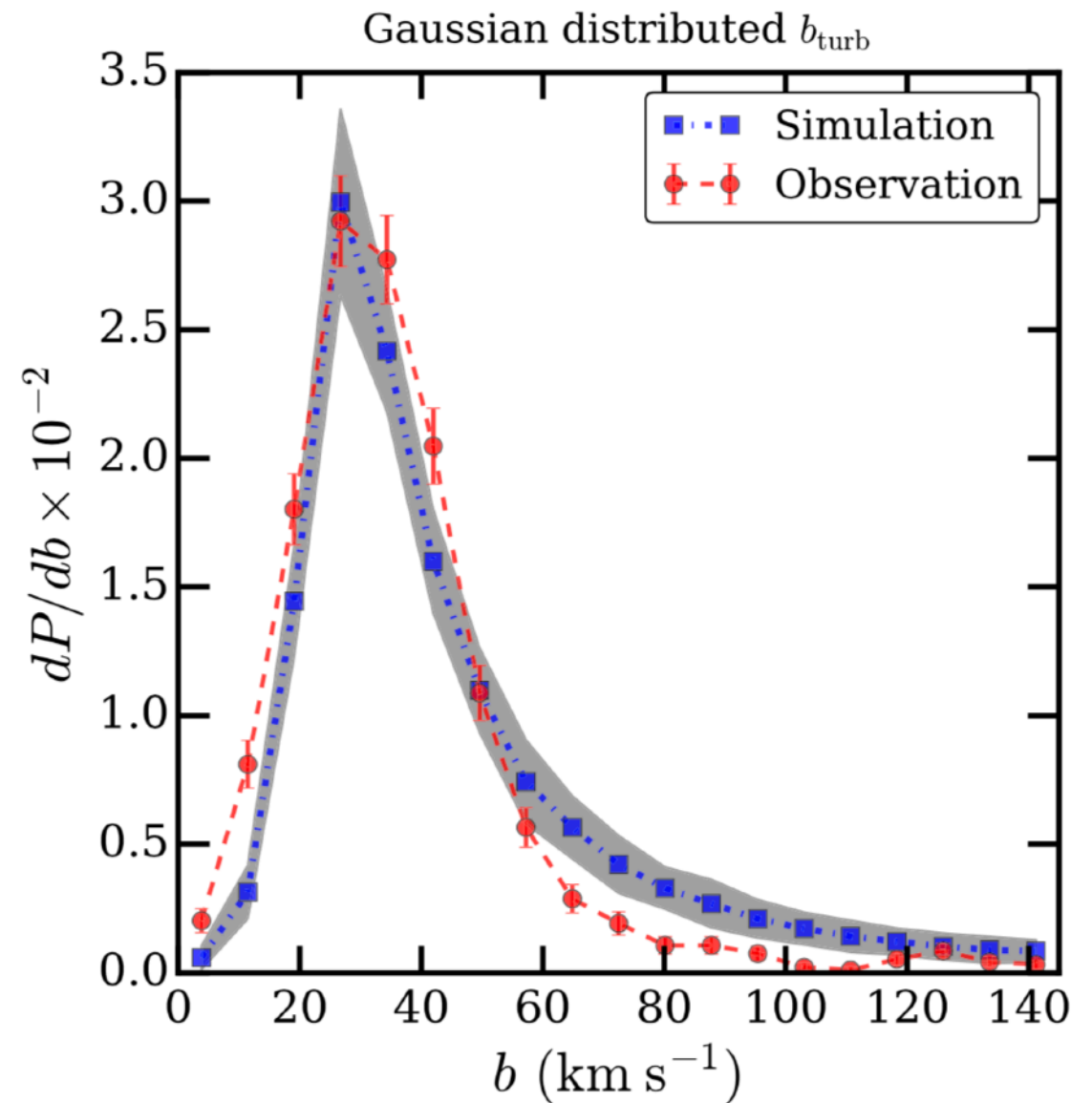
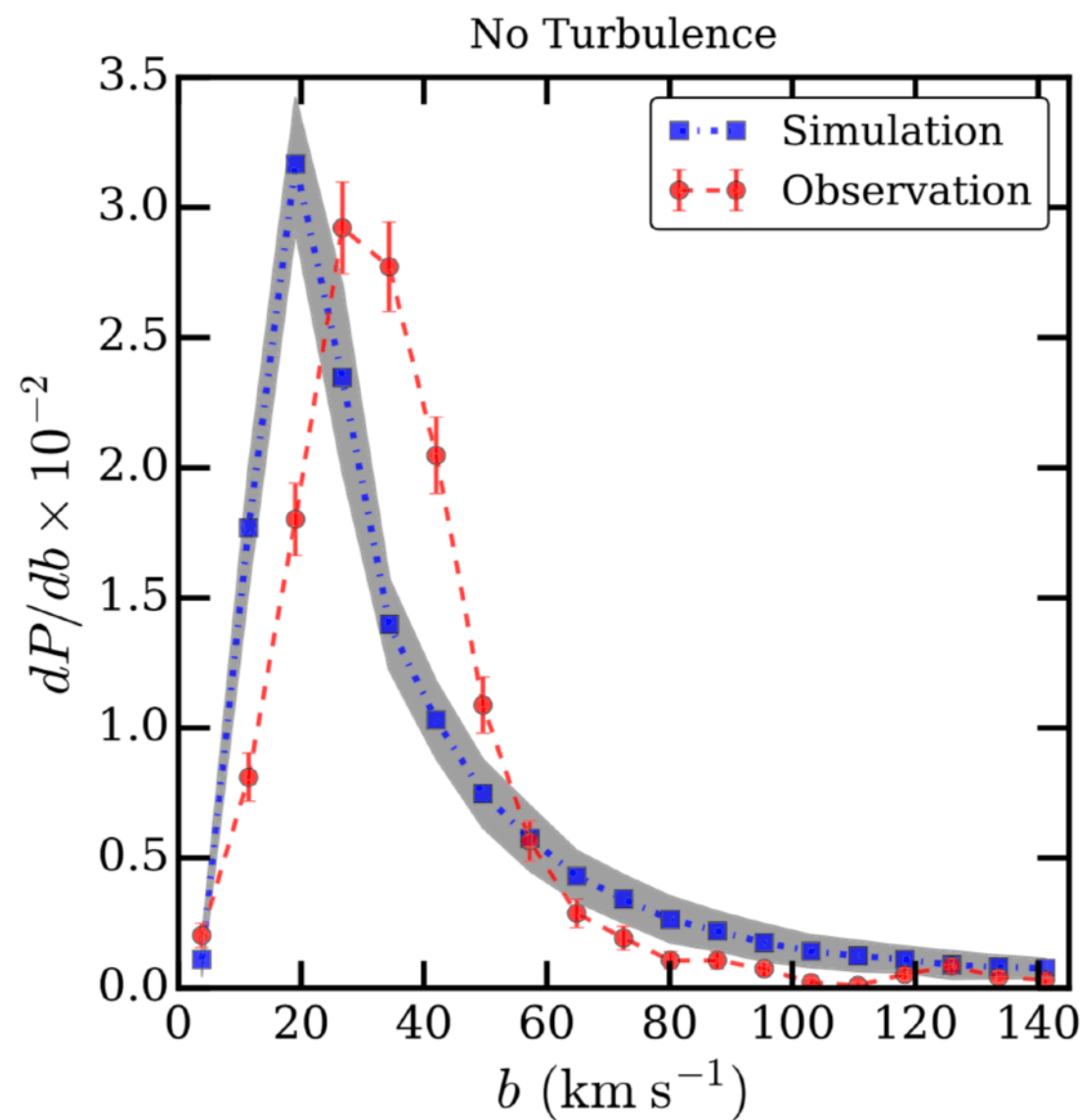
e.g. Madau & Efstathiou (1999)

Compton heating by X-rays?



Compton heating effective at $\Delta < 1$ and $z < 2$ (heating scales with Δ instead of Δ^2 for photo-heating): Ly- α forest at $z=0.1$ largely unaffected.

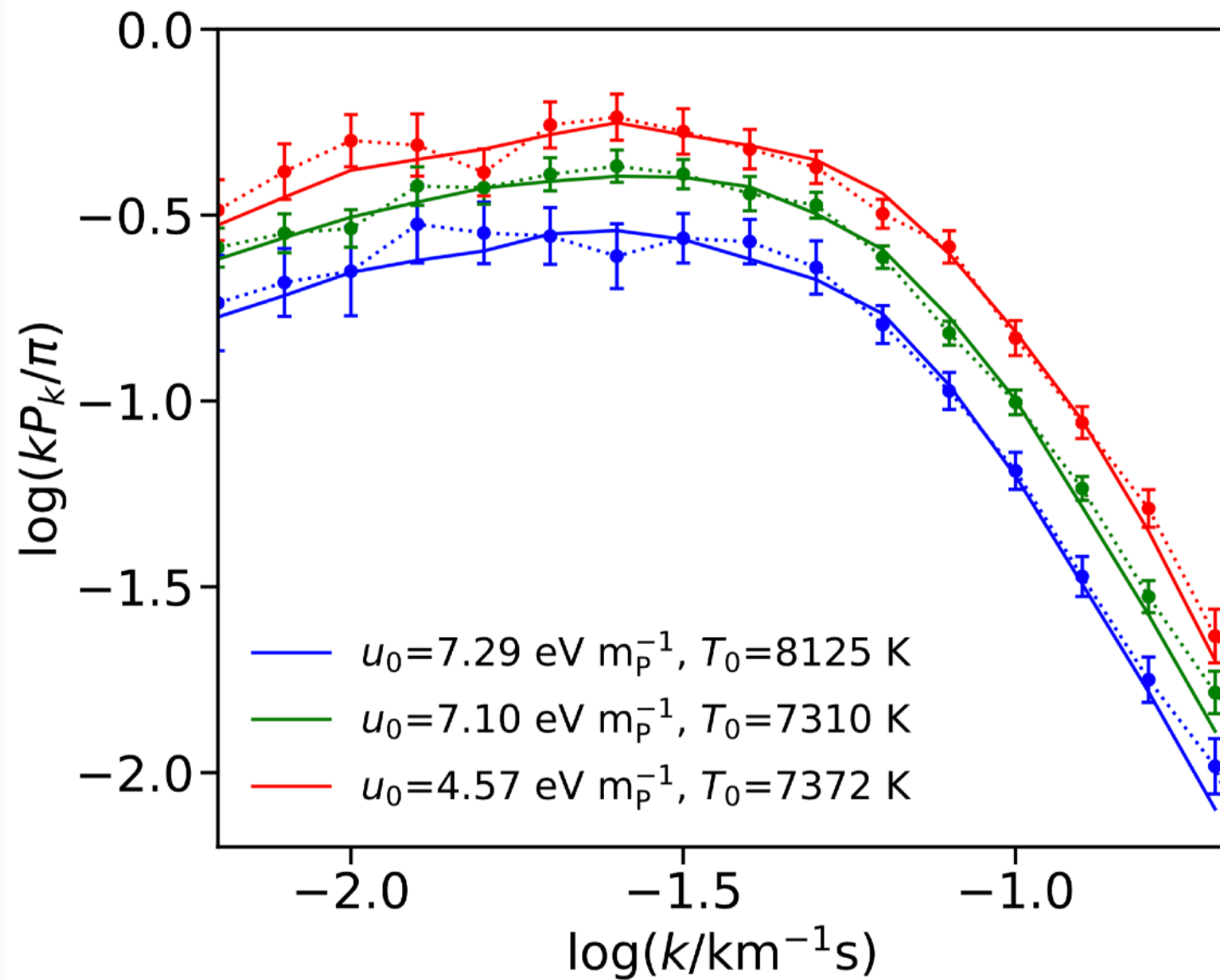
Unresolved turbulence?



Gaikwad et al. (2017)

see also Oppenheimer+09, Evoli+11, Iapichino+13

IGM thermal state at $z \sim 5$



Boera et al. (2019)

- Instantaneous reionisation model gives $z_{\text{rei}} \sim 8.5$ for best fit thermal parameters ([Elisa's talk](#))
- See also [Irsic et al. \(2017\)](#) for WDM/fuzzy DM limits from $P(k)$ at $z \sim 5$



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George Becker (UCR)

James Bolton (Nottingham)

Jonathan Chardin (Strasbourg)

Martin Haehnelt (Cambridge)

Vid Irsic (UW)

Laura Keating (CITA)

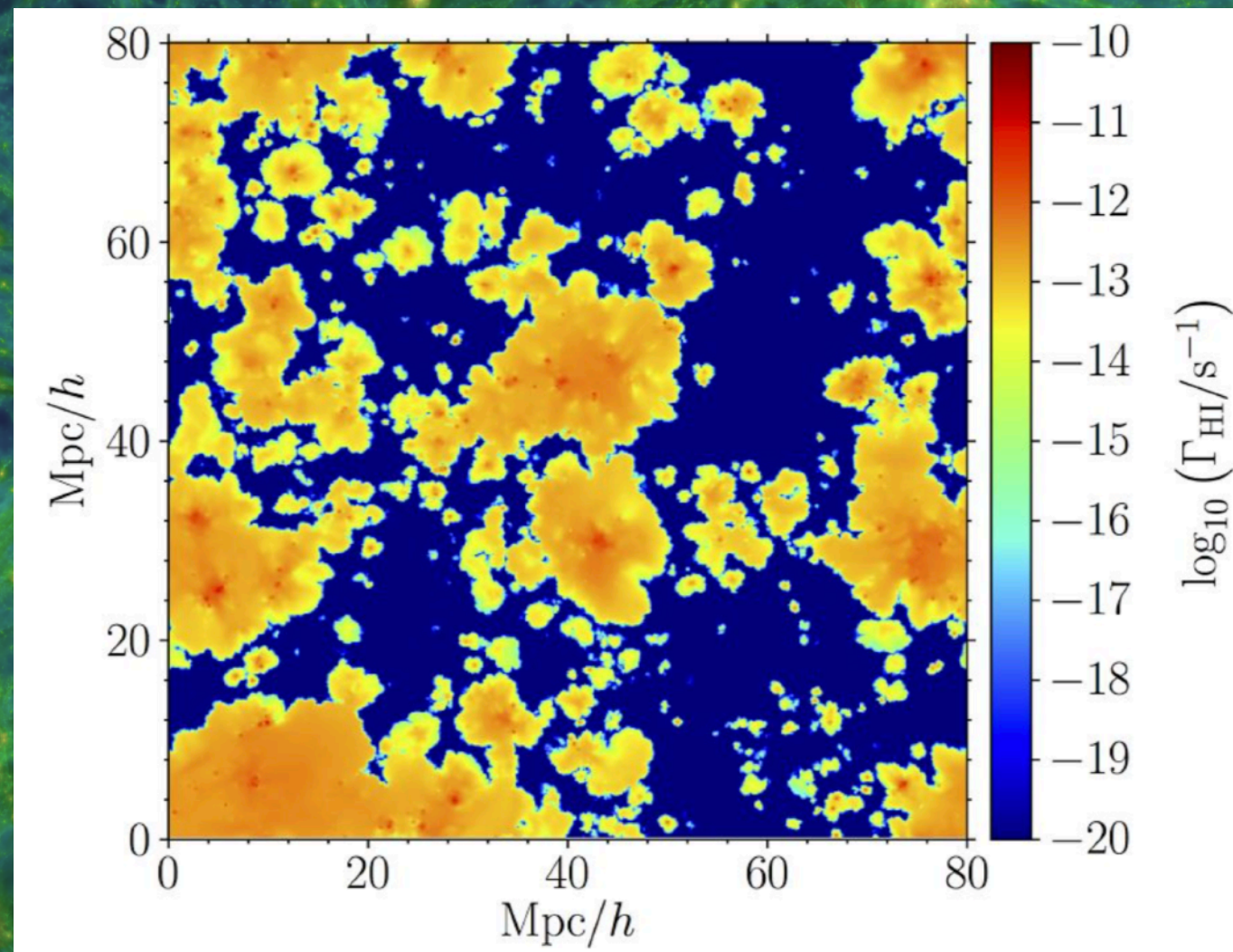
Girish Kulkarni (TIFR, Mumbai)

Avery Meiksin (Edinburgh)

Debora Sijacki (Cambridge)

Matteo Viel (SISSA)

- >100 hydrodynamical IGM simulations with P-Gadget-3, building on Sherwood ([Bolton+17](#));
- Runs are coupled to the output of empirically calibrated RT simulations (ATON) to model patchy ionisation and heating



Summary

- Lyman- α forest ($N_{\text{HI}} \sim 10^{13} - 10^{14} \text{ cm}^{-2}$) at $z \sim 0.1$
 - Hydro-simulations struggle to reproduce line widths at $z=0.1$;
 - Reproducing line width distribution with additional photo-heating requires (unphysically) hard UVB spectrum;
 - AGN/stellar feedback has minimal impact on line widths for models examined — need gas at just the right temperature and density;
 - Compton heating by X-rays also no help — heats low density gas;
 - Unresolved turbulence? Or something else?
- Lyman- α forest at $z > 5$
 - New, self-consistent modelling of emissivity and patchy reionisation in IGM hydro-simulations; next steps are matching with $P(k)$.